When the height $h$ = 150cm, the weight is $w = 49kg$ (approximately)
Therefore, the weight of the person whose height 150cm is 49 kg.

**Example 11.13**

A tree is growing so that, after t - years its height is increasing at a rate of $\frac{18}{\sqrt{t}}$ cm per year. Assume that when $t$ = 0, the height is 5 cm.

(i) Find the height of the tree after 4 years.

(ii) After how many years will the height be 149 cm?

**Solution**

The rate of change of height $h$ with respect to time $t$ is the derivative of $h$ with respect to $t$.

$$\text{Therefore, } \frac{dh}{dt} = \frac{18}{\sqrt{t}} = 18t^{-\frac{1}{2}}$$

So, to get a general expression for the height, integrating the above equation with respect to $t$.

$$h = \int 18t^{-\frac{1}{2}}dt = 18(2t^{\frac{1}{2}}) + c = 36\sqrt{t} + c$$

Given that when $t$ = 0, the height $h$ = 5 cm.

$$5 = 0 + c \Rightarrow c = 5$$

$$h = 36\sqrt{t} + 5.$$

(i) To find the height of the tree after 4 years.

When $t$ = 4 years,

$$h = 36\sqrt{t} + 5 \Rightarrow h = 36\sqrt{4} + 5 = 77$$

The height of the tree after 4 years is 77 cm

(ii) When $h$ = 149cm

$$h = 36\sqrt{t} + 5 \Rightarrow 149 = 36\sqrt{t} + 5$$

$$\sqrt{t} = \frac{149 - 5}{36} = 4 \Rightarrow t = 16$$

Thus after 16 years the height of the tree will be 149 cm.

**Example 11.14**

At a particular moment, a student needs to stop his speedy bike to avoid a collision with the barrier ahead at a distance 40 metres away from him. Immediately he slows (retardation) the bike under braking at a rate of 8 metre/second$^2$. If the bike is moving at a speed of 24m/s, when the brakes are applied, would it stop before collision?



**Solution**

Let $a$ be the acceleration, $v$ be the velocity of the car, and $s$ be the distance.



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